Breast cancer survival Prediction (Project 2)

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Table of Contents

01. **Background Information**

02. **The dataset I used**

03. **Methodology**

04. **Analysis**

05. **ML pipeline**

06. **Conclusion / Room for Improvement**

Background Information

About breast cancer

Breast cancer is the most commonly found cancer in women across the globe. It is especially dangerous in low-income countries, where treatment is not readily available and affordable. In fact, it is the #1 cause of mortalities in over 100 countries.

Hypoxia is a condition where body tissues lack the sufficient oxygen they need to function. Symptoms range from restlessness to being placed in a coma.

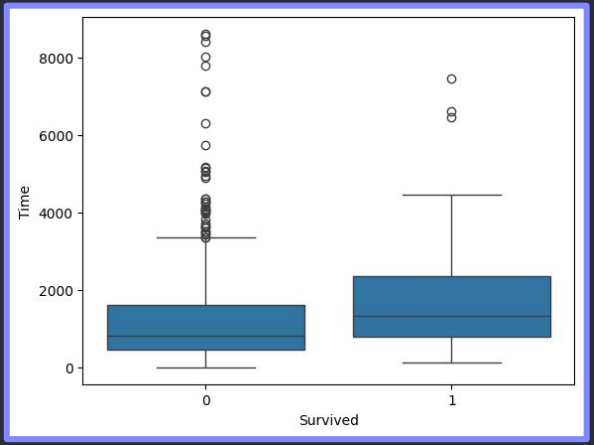
So, how do the two conditions relate? Medical research shows that hypoxia in breast tumors promote the formation of new blood vessels from old ones, and also cause metabolic changes, aiding cancer cells from evading our immune systems, and also potentially leading to the formation of new tumors in different parts of the body, in a process known as metastasis.

Although hypoxia is not a central focus of my project, it is still important to understand how it affects survival.

About the Dataset

Two datasets were used: One containing the survival data of patients and the other containing gene expression data of the same patients. Some basic statistics of the datasets are listed below. Both datasets were from TCGA.

- Roughly around ~83% of the selected samples in the dataset survived.
- The mean survival time was 1269 days, with extremes of 0 and 8605 days (refer to chart on the right)
- The dataset consisted of 1235 samples, and 1214 samples were shared across both datasets.
- The dataset contained over 20000 genes, in which I had to select only the most statistically significant ones



Analysis (Preprocessing)

Step 1 - Selection of Significant Genes

1. This step involved removing genes which had a low expression (represented by a numerical value < 1) in 80+% of the dataset.

   This led to a reduction in the genes of the dataset from ~20000 to ~17000, which effectively cut 15% of the total genes.

2. Then, I conducted a log-rank test, which examines if genes have a significant effect on survival. For my p-value threshold, I took 0.01, so only the most statistically significant genes were taken. This led to a whopping 89.5% of genes being cut, leaving only 1850 left.

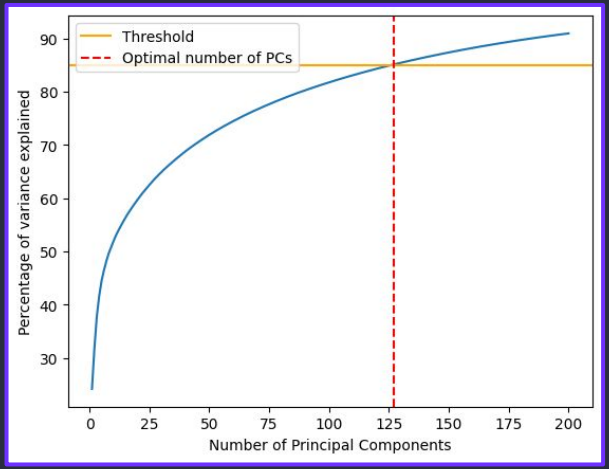
3. Finally, I ran the remaining genes through a Cox Proportional Hazards Fitter, which creates a Cox Proportional Hazards (CoxPH) model for every gene expression, and then tests the statistical significance of the gene. After the filter, I was left with 989 genes to finally work with, which was roughly a 46.3% reduction.

Step 2 - Preprocessing of Data

The main process used in this stage was Principal Component Analysis (PCA), a method used to reduce dimensionality by grouping features together into principal components (PCs). I selected a variance threshold of 85%, as it effectively results in an optimal number of PCs which explain most of the variance of the dataset.

Following PCA, I had only 127 features to work with, as opposed to the previous 989, considerably reducing the number of features.

After PCA, I merged both datasets by only selecting samples included in both sets, which resulted in a final merged set consisting of 1214 samples.



Modeling and Risk Group Stratification

Survival Prediction Using CoxPH and RSF Models

Splitting Of Test Data

My initial idea was to use a simple train test split approach where 20% of the data would be used for testing and the rest for training. However, this resulted in a large range of C-index values, which proved to be unreliable. This led to me changing the methodology to a KFold method, where the data is split into training in testing 5 different times, so that all of the data is used for testing at some point. I then took the averages of the C-indexes produced by the different splits of train and test data, and found the standard deviation to product a reliable result of the C-index for each model.

Cox Proportional Hazards Model

I used sksurv's CoxPHSurvivalAnalysis model in order to analyse and group patients into different risk groups. After the KFold process was complete, I checked the C-index average and standard deviation. The results are as below:

Mean: 0.694
Standard Deviation: 0.045

The results portray an acceptable model which decently performs at predicting the probability of an event occurring for a patient.

Small Fact: After the risk group stratification, I found out that ~68.5% of patients were classified in the same risk group by both models

Random Survival Forest

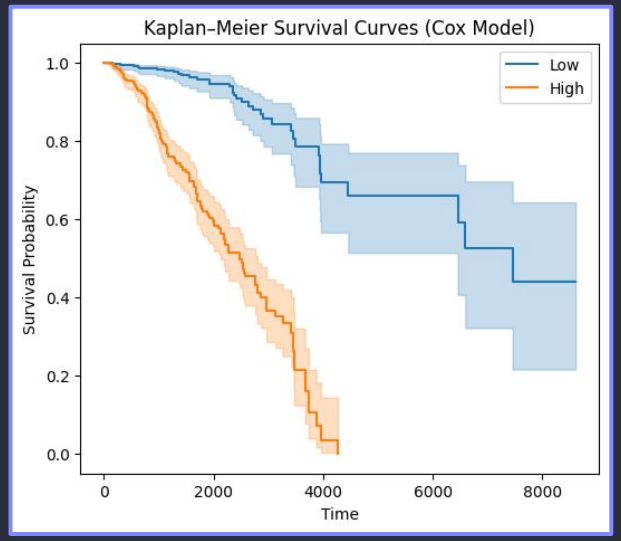
I initially ran a grid search with cross validation in order to fine tune hyperparameters and find the optimal parameters for my model. After training the model and calculating C-index metrics, the results came out as follows:

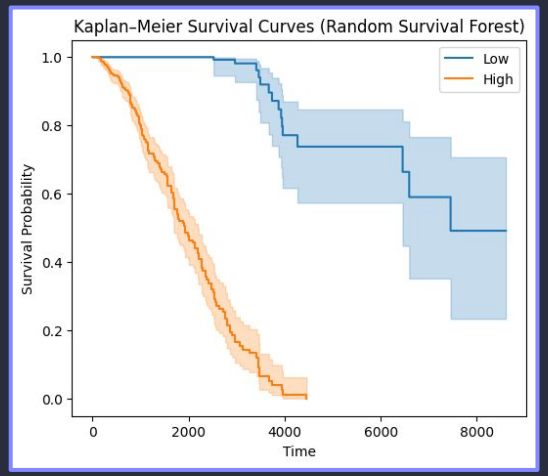
Mean: 0.701
Standard Deviation: 0.039

The results were better than the CoxPH model, as most C-index values larger and were within a smaller range away from the mean than in the CoxPH model.

Kaplan-Meier Curves

Kaplan-Meier Curves are used to represent how chances of survival changes over time, with a certain condition being met. In my KM curves, I decided to model the relationship between survival chance and time given the predicted risk group. Ideally, the low risk group curve should be above the high risk group one, as over time patients in the high risk group are more likely to experience an event. Furthermore, the high risk group curve should have a steeper fall than the low risk group curve, as survival sharply decreases as time passes for patients at high risk. As you can see by the charts, both these constraints are met by both models, symbolising models which can accurately differentiate patients between high and low risk groups.



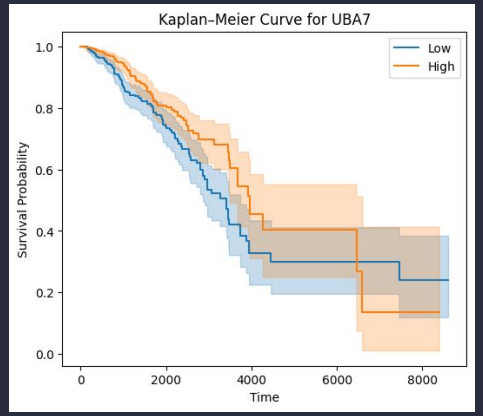


Key Genes and Pathway Analysis

Key Genes Affecting Survival Rates

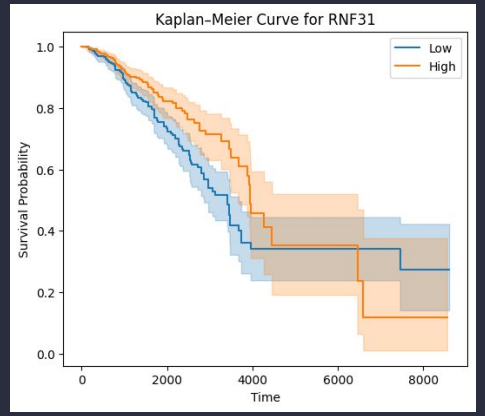
1. UBA7

Research shows that the UBA7 gene helps trigger anti-tumor immunity, aiding in the protection against breast cancer. If this gene has a low expression, it could cause worse patient prognosis and lead to advanced stages of cancer.



2. RNF31

Research shows that the RNF31 gene acts as a neutral gene, both promoting tumor growth as well as acting as a tumor suppressor. This is shown in the KM curve for the gene, as we can see that a high expression improves survival chances at first but ultimately acts neutrally.



Pathway Analysis Results

After conducting pathway analysis, there were many pathways where the specified genes were involved. Here are some examples of some of the pathways:

1. Interferon Alpha
2. Fatty Acid Metabolism
3. Interferon Gamma
4. Apoptosis
5. TGA-Beta

Amongst many more

I will now explain 1 of the pathways in detail

1. Interferon Alpha - An innate antiviral layer of protection, which activates other pathways and is involved in stimulating immune responses in cancer. Some examples of genes in this pathway are: UBA7, TAP1, and IFI35. Using this information, we can loosely infer a positive association between the expression of genes in this pathway and survival rates. This evidence is supported by the KM plot of UBA7 in the previous slide, as a higher expression typically resulted in a higher rate of survival over time.

Possible Improvements

Mismatch in the Survival Data

Out of the samples chosen, close to 83.7% of them survived. I believe the models I trained can perform much better if I had imputed more samples of patients who had passed away to smooth out the disparity in the survival, as the model may overfit to cases where the patient survives due to the lack of samples where the patient experienced an event

Improvements in RSF Tuning

I believe that the Random Survival Forest could perform much better if given a large parameter grid during the grid search. Although not possible with the resources I was running the code with, I still think there would be a noticeable and significant difference in the models performance given optimal hyperparameter tuning, as it ensures the model learns relationships to its best ability.